

Correct Semantics for `while`

$$(W\text{-}WHILE) \frac{}{\langle \textit{while } B \textit{ do } C, s \rangle \rightarrow_c \langle \textit{if } B \textit{ then } (C; \textit{while } B \textit{ do } C) \textit{ else skip}, s \rangle}$$

All this rule does is ‘unfold’ the while loop once. If we could write down the infinite unfolding, there would be no need for the `while` syntax.